

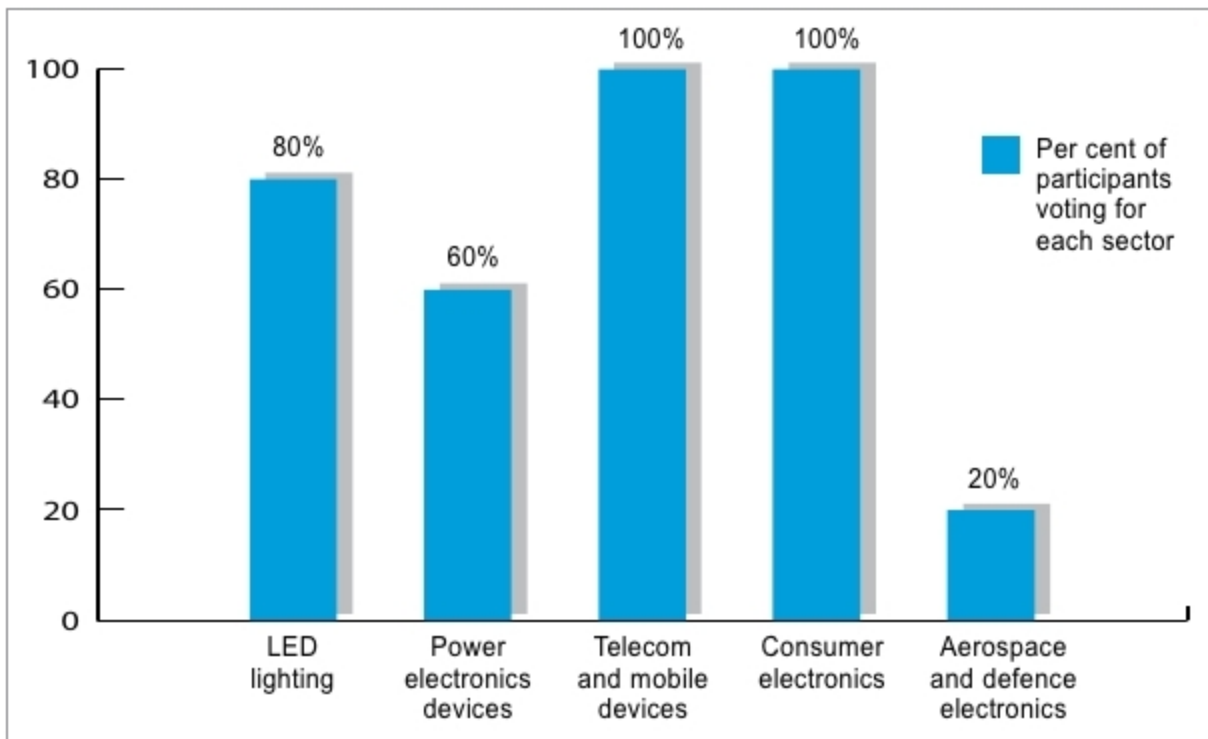


Fig. 1: Views about the importance of major growth driving sectors

Manufacturing partnerships in these three segments have been growing steadily, as market demand for the products is extremely high. OEMs are also striving to cut costs to maintain their competitive advantage in the face of rapidly changing market conditions, technological advances and global competition.

There are further opportunities for EMS providers outside these three segments, namely, in power and strategic electronics (including aerospace, defence and railways). With increasing demand for the transition of energy segment from electrical to electronic, power electronics is emerging as one of the most exciting areas in the industry today.

Aerospace and defence (A&D) OEMs have been increasingly depending on EMS providers to address risk management, logistics and after-market service needs. EMS companies with global supply chains and advanced technological capabilities have the best proposition to exploit this trend.

Success mantras

EMS companies operate as strategic partners of OEMs by providing them with a full range of services. These include contract design service, prototyping, final system assembly, configuration, order fulfilment and after-market services, including repair. By using the services of EMS providers, OEMs can concentrate on their core

competencies such as research and product development, brand building, sales and marketing.

Outsourcing to EMS providers also enables OEMs to gain access to the latest equipment, process knowledge and manufacturing know-how, without having to make substantial capital investments, as the risks are converted into variable costs.

Ever-increasing end-user demands and fast-paced technological developments compel OEMs to continuously introduce new and innovative products in the electronics market. Consequently, OEMs have to depend

increasingly on EMS providers, who offer significant benefits such as cost savings, reduced time-to-market, reduced time-to-volume, quality and flexibility.

Strategic partnerships with OEMs can enable EMS providers as well as OEMs to contribute effectively to the success of the end product. This, in turn, helps both parties achieve high profitability and a good market share.

Since OEMs are very selective in choosing their EMS partners, the latter need to focus on nurturing long-term relationships with their customers through enhanced value-added services, strategic partnerships and alliances, as well as through diversification.

In response to the growing competition in the industry, EMS providers continuously adopt innovative and strategic business models. These include better knowledge of customer needs, understanding the business models of customers/OEMs, effective communication tools, creating global footprints and focusing on core competencies.

According to the survey participants, the top three business strategies (Fig. 2) to get a competitive advantage while acquiring business are:

1. Providing value-added services
2. Effective supply chain management

Low-mix, high-volume or high-volume, low-mix?

Indian EMS providers tend to show long-term affinity towards a low-mix, high-volume module, which is a contract manufacturing setup where there are few assemblies produced in large quantities. Such a production arrangement may last for weeks or even months using the same setup. Benefits of this technique are that changeovers can be kept at minimum levels and equipment utilisation rates are significantly high.

Contract manufacturers have proved to be more efficient when running at high volumes, which require minimal engineering changes. A low-mix, high-volume process ensures economies of scale, higher productivity due to minimum changeover time and optimum machine utilisation. Considering the low margin and highly competitive price scenario, optimisation of supply chain cost is crucial to the success of this model.

While the low-mix, high-volume method of production is adopted for EMS companies catering to the mass manufacturing segment, recent industry trends show a shift towards a high-mix, low-volume production process, which puts high focus on quality and customisation as per customer requirements.

Considering a high-margin and niche market scenario, even major changes in market dynamics often do not impact such a production process heavily. This kind of production arrangement mostly caters to the strategic electronics sector, and emphasis is on faster throughput and a high-quality finished product. OEMs that prefer such solutions are willing to pay a higher price without compromising on quality. However, in this model, success also depends on controlling and improving the supply chain efficiently.